Kaushlendra Kumar

Curriculum Vitae

School of Mathematical Sciences
Queen Mary University of London
✉ *kaushlendra.kumar@qmul.ac.uk*
k-kumar.netlify.app/
ORCID *arXiv* *InspireHEP*

Profile

Research	My research lies in mathematical and theoretical physics, connecting exact gauge fields, quantum/noncommutative geometry, spectral distance, quantum information geometry, and emergent spacetime. Recent work computes black-hole quantum-geometric flow and phase diagnostics, finite spectral distances with quantum-information meaning, exact Yang–Mills/coset fields and source/transport laws, finite Clifford spectral triples, and matrix-model corrections to gravitational dynamics.
Keywords	Quantum geometry, exact Yang–Mills fields, noncommutative geometry, spectral distance, quantum information geometry, black-hole transport, matrix/emergent gravity.
Identifiers	ORCID: 0000-0001-5665-575X; arXiv: kumar_k_2; InspireHEP: 1706741.
Publication summary	11 refereed journal articles, 2 conference proceedings, and 5 current public preprints.
Research website	k-kumar.netlify.app/work: public research map, papers/preprints, talks, public engagement, and updated programme status.

Positions

Mar 2024–Nov 2027	**DFG Walter-Benjamin Research Fellow**, *School of Mathematical Sciences*, Queen Mary University of London, Quantum Riemannian Geometry with Shahn Majid.
Jul 2023–Feb 2024	**Postdoc**, *Institute of Physics*, Humboldt University of Berlin, Germany. Funded by the Volkswagen Stiftung grant of Nichol Furey.
5 Mar 2023–15 Jun 2023	**Junior Research Fellow**, *Erwin Schrödinger Institute for Mathematics and Physics*, University of Vienna.
May 2022–Mar 2023	**Postdoctoral research**, *Leibniz University Hannover*, DAAD-supported, including salaried employment October–December 2022.

Major Fellowships

2024–2027	**Deutsche Forschungsgemeinschaft (DFG)**, Walter-Benjamin Fellowship, Queen Mary University of London.
2018–2022	**Deutscher Akademischer Austauschdienst (DAAD)**, Doctoral scholarship, Leibniz University Hannover.
2016–2018	**Council for Scientific and Industrial Research (CSIR)**, Junior Research Fellowship (JRF), IISER Kolkata.

Education

Oct 2018–May 2022	**PhD, Theoretical Physics**, *Leibniz University Hannover*, Germany, Magna cum Laude. Defended 4 May 2022.
Thesis	Solutions of Yang–Mills theory in four-dimensional de Sitter space.
Jul 2015–May 2018	**MSc, Physics**, *IISER Kolkata*, India, CGPA: 9.25/10. Thesis defended 20 July 2017.
Jul 2010–May 2015	**BTech & MTech, Biochemical Engineering**, *IIT (BHU), Varanasi*, India, CGPA: 7.88/10. Thesis defended 27 May 2015.

List of publications

(a) Refereed journal articles

2026 **K. Kumar**. Characteristic lightcone sources in SO(1,3) Yang–Mills theory. *Phys. Lett. B* **880**, 140865 (2026). doi:10.1016/j.physletb.2026.140865.

2024 **K. Kumar** and H. C. Steinacker. Modified Einstein equations from the 1-loop effective action of the IKKT model. *Class. Quantum Grav.* **41**, 185007 (2024). doi:10.1088/1361-6382/ad6e4b.

2024 S. Hirpara, **K. Kumar**, O. Lechtenfeld, and G. Picanço Costa. Exact gauge fields from anti-de Sitter space. *J. Math. Phys.* **65**, 072903 (2024). doi:10.1063/5.0150027.

2022 **K. Kumar**, O. Lechtenfeld, G. Picanço Costa, and J. Röhrig. Yang–Mills solutions on Minkowski space via non-compact coset spaces. *Phys. Lett. B* **835**, 137564 (2022). doi:10.1016/j.physletb.2022.137564.

2022 **K. Kumar**, O. Lechtenfeld, and G. Picanço Costa. Trajectories of charged particles in knotted electromagnetic fields. *J. Phys. A: Math. Theor.* **55**, 315401 (2022). doi:10.1088/1751-8121/ac7c49.

2022 L. Hantzko, **K. Kumar**, and G. Picanço Costa. Conserved charges for rational electromagnetic knots. *Eur. Phys. J. Plus* **137**, 407 (2022). doi:10.1140/epjp/s13360-022-02563-4.

2021 **K. Kumar**, O. Lechtenfeld, and G. Picanço Costa. Instability of cosmic Yang–Mills fields. *Nucl. Phys. B* **973**, 115583 (2021). doi:10.1016/j.nuclphysb.2021.115583.

2020 **K. Kumar** and O. Lechtenfeld. On rational electromagnetic fields. *Phys. Lett. A* **384**, 126445 (2020). doi:10.1016/j.physleta.2020.126445.

2018 **K. Kumar** and B. Chakraborty. Spectral distances on the doubled Moyal plane using Dirac eigenspinors. *Phys. Rev. D* **97**, 086019 (2018). doi:10.1103/PhysRevD.97.086019.

2018 Y. C. Devi, **K. Kumar**, B. Chakraborty, and F. G. Scholtz. Revisiting Connes' finite spectral distance on noncommutative spaces: Moyal plane and fuzzy sphere. *Int. J. Geom. Methods Mod. Phys.* **15**, 1850204 (2018). doi:10.1142/S0219887818502043.

2015 **K. Kumar**, S. Prajapat, and B. Chakraborty. On the role of Schwinger's SU(2) generators for Simple Harmonic Oscillator in 2D Moyal plane. *Eur. Phys. J. Plus* **130**, 120 (2015). doi:10.1140/epjp/i2015-15120-7.

(b) Refereed conference proceedings

2023 **K. Kumar**. SO(1,3) Yang–Mills solutions on Minkowski space via cosets. *SciPost Phys. Proc.* **14**, 033 (2023). doi:10.21468/SciPostPhysProc.14.033.

2023 **K. Kumar**. On Yang–Mills fields from anti-de Sitter spaces. *J. Phys. Conf. Ser.* **2667**, 012018 (2023). doi:10.1088/1742-6596/2667/1/012018.

(c) Preprints

2026 **K. Kumar**. Helstrom geometry with time-dependent spectral calibration. arXiv:2610.33848 [quant-ph]; prepared for the XLIII Workshop on Geometric Methods in Physics proceedings.

2026 **K. Kumar**. Laughlin quasihole geometry from a single snapshot ensemble. arXiv:2608.22384 [quant-ph].

2026 **K. Kumar**. Calibrated Helstrom geometry on the Bloch ball via Connes spectral distance. arXiv:2606.13824 [quant-ph].

2026 **K. Kumar** and S. Majid. Geometric Aharonov–Bohm phase effect around a black hole. arXiv:2605.28582 [gr-qc].

2026 **K. Kumar** and S. Majid. Geodesic flows on a black-hole background. arXiv:2603.03222 [gr-qc].

Professional service

Referee — *Journal of Geometry and Physics*; *International Journal of Theoretical Physics*.

Teaching

Student collaboration — Helped master's students Savan Hirpara and Jona Röhrig with calculations on exact-field projects guided by Olaf Lechtenfeld at Hannover.

Spring, 2020 — **Introduction to Quantum Theory**, *intensive 1-week revision course*, ITP-LUH.

Fall, 2019 — **Mathematical Methods of Physics**, *intensive 1-week revision course*, ITP-LUH.

Spring, 2019 — **Lie groups, Algebra and their Representations**, *teaching assistant*, ITP-LUH.

Fall, 2017 — **Advanced Mathematical Methods of Physics**, *teaching assistant*, IISER-K.

Spring, 2017 — **Intermediate Electricity and Magnetism**, *teaching assistant*, IISER-K.

Selected talks

1 Jul, 2026 — **XLIII Workshop on Geometric Methods in Physics, Białystok**.
Scalar-anchored Connes distance and Helstrom geometry of the qubit Bloch ball (conference talk).

9 Dec, 2025 — **Quantum Algebras seminar, Queen Mary University of London**.
Geodesic flows on a black-hole background.

27 Jul, 2023 — **QTS12 conference, Prague**.
Exact gauge fields from anti-de Sitter space.

4 May, 2023 — **Erwin Schrödinger Institute, Vienna**.
Exact gauge fields from anti-de Sitter space.

18 Jul, 2022 — **GROUP34 conference, Strasbourg**.
Yang–Mills solutions on Minkowski space via non-compact coset spaces.

Selected schools and conferences

15–17 Jul 2026 — **Black Hole Horizons and Beyond**, Durham, United Kingdom.

29 Jun–4 Jul 2026 — **XLIII Workshop on Geometric Methods in Physics**, Białystok, Poland.

7–11 Jul 2025 — **A Quantum of Noncommutativity**, Ambleside, United Kingdom.

2020–2022 — **Saalburg Summer Schools**, Quantum field theory, integrable systems and gravitation, Germany.

2–7 Aug 2021 — **International Congress on Mathematical Physics**, Geneva, Switzerland.

23–27 Sep 2019 — **Special Holonomy – Geometry and Physics**, Freiburg, Germany.

Language skills

Languages — Maithili, Hindi and Bengali; fluent English; German B1 (DAAD-funded course at Carl Duisberg Centrum, Berlin, June–September 2018).

Research computation

Tools — Mathematica/Wolfram Language, LaTeX, GnuPlot, symbolic and numerical computation.

Methods — Finite-dimensional spectral-geometry calculations, exact tensor/curvature computations, Yang–Mills mode algebra, PDE/transport postprocessing.

Systems — Linux, macOS, Windows.